

Questions on GANs and Recap of Previous Generative Models

VAEs

1. What is the reparametrization trick?
2. What is a VAE? (according to slide 58)
3. Why is it called an autoencoder?
4. What variations of VAEs have we seen?
5. What is the idea of VQVAE?
6. Which other idea we have seen earlier in this class does VQVAE use?

GANs

1. What is the core idea of GANs? (Building a sampler..., what are the requirements?)
2. What is the objective of the original GAN paper (slide17)? What is its training procedure?
3. How can we evaluate GANs?
4. What is the inception score?
5. What is the FID?
6. Why is FID a better measure than IS?
7. What does the Discriminator aim for?
8. What does the generator aim for?
9. Why have several distribution measures been discussed in this context?
10. What do we mean by "mode collapse" in the context of GANs?
11. How can discriminator saturation be avoided?
12. Describe the DCGAN architecture.
13. What was good about DCGAN?
14. How does Progressive Growing GAN improve DCGAN?
15. What is SAGAN?
16. Describe the idea of the BigGAN architecture. What is BigGAN good at?
17. Describe the StyleGAN architecture. What is StyleGAN good at?
18. Which of the discussed architectures can produce good, high-resolution images?